

Supplemental Material for “RoCBF-Net: A Tunable Safety Architecture for High-Order CBFs Under Uncertainty—GP Correction, Tunable Margins, and the Deployment Envelope”

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Supplemental Material

Reference numbers follow the numbering of the main manuscript.

S1 Evidence Base and Result Inventory

The Measurement and Control submission uses the 5th-order Phase 5 result inventory as the primary evidence base. The root-level inventory in `results/phase5/` contains the complete fair-comparison matrix: 8 methods, 8 conditions, and 5 seeds, for 320 JSON result files. The strict inventory audit reports 320/320 matched files, no missing root-level files, no extra root-level JSON files, and no JSON parsing errors.

The older Phase 4 3rd-order results are retained only as historical and auxiliary material. Their inventory is intentionally not used as the primary M&C evidence base because it is incomplete for PPO-GP-HOCBF on several conditions. Where Phase 4 is mentioned in the manuscript, it is used only to identify the previously observed generic-GP and out-of-distribution failure mode, not to support the main 5th-order safety comparison.

S2 Full ϵ_κ Sensitivity Data

Table S1 reports the numerical data behind the kappa-sensitivity figure in the main manuscript. Values are mean CBF violation rates over the available kappa-sweep seeds. Unless otherwise noted, each cell uses three seeds.

Table S1: ϵ_κ sensitivity sweep on three representative 5th-order CCS scenarios. Violation rates are percentages; values in parentheses report the number of seeds when fewer than three completed results are available.

Scenario	$\kappa = 0$	$\kappa = 0.1$	$\kappa = 0.3$	$\kappa = 0.5$	$\kappa = 1.0$
S2: Pressure	0.00	0.00	0.00	99.36	99.42
S3: Coupled	33.47	0.00	0.10	0.25	40.30 (2)
S4: Nonlinear	0.00	0.00	0.00	0.10 (2)	46.83 (2)

These data support three practical conclusions. First, additive perturbations (S2 and S4) require no robustness margin in the tested regime; GP mean correction alone is sufficient. Second, state-dependent coupling (S3) requires a small positive margin: $\kappa = 0$ fails, while $\kappa = 0.1$ restores zero observed violation. Third, the full theoretical margin $\kappa = 1.0$ is not the empirical optimum in any tested scenario and can collapse the feasible set.

S3 Coupling-Strength Deployment Envelope

Table S2 gives the data behind the main manuscript coupling-gradient figure. Each cell is the mean CBF violation rate over two seeds, with 50 episodes and 500 steps per episode.

Table S2: S3 coupling-strength gradient: violation rate (%) versus ϵ_κ . Safe cells are those below 1% violation.

Coupling strength	$\kappa = 0$	$\kappa = 0.1$	$\kappa = 0.3$	$\kappa = 0.5$	$\kappa = 1.0$
$\gamma = 0.5$	8.32	0.00	0.00	0.19	0.56
$\gamma = 1.0$	33.47	0.00	0.10	0.25	40.30
$\gamma = 1.5$	0.51	75.26	94.69	94.84	97.28
$\gamma = 2.0$	0.19	98.92	99.83	99.98	99.98

The safe ϵ_κ interval narrows with coupling strength and collapses to $\{0\}$ once the perturbation amplification rate reaches $\gamma \geq 1.5$. In that regime, adding margin primarily tightens an already difficult QP; infeasibility becomes a diagnostic signal for fallback rather than a reason to increase ϵ_κ .

S4 Perturbation Magnitude Sweep

Table S3 reports the S1 heat-absorption magnitude sweep for GP-HOCBF with $\epsilon_\kappa = 0$. This experiment isolates the GP mean-correction envelope without adding a robustness margin.

Table S3: Perturbation magnitude sweep for GP-HOCBF ($\epsilon_\kappa = 0$). Violation and intervention rates are percentages.

Magnitude	Δf_h (kJ/kg)	CBF violation	QP intervention
Mag10	-10	0.0	100.0
Moderate	-15	0.0	100.0
Mag25	-25	0.0	100.0
Mag50	-50	0.0	100.0
Mag75	-75	100.0	100.0
Mag100	-100	100.0	100.0

The GP-only configuration remains safe through Mag50 and fails sharply at Mag75 and above, supporting the deployment-envelope interpretation in the main manuscript.

S5 Load-Following and Constraint Decomposition

The load-following AGC condition is included in the complete Phase 5 inventory but omitted from the main safety table to keep that table focused on static perturbation scenarios. Table S4 reports the corresponding mean CBF violation rates.

Table S4: Load-following AGC condition: mean CBF violation rate (%) over 5 seeds.

Method	CBF violation
PPO	0.08
PPO-Lagrangian	0.08
PPO-CBF	0.08
PPO-HOCBF	0.00
PPO-GP-HOCBF	0.00
PPO-RHOCBF	0.00
RoCBF-Net	0.00
NMPC	0.00

Per-constraint logs show no observed pressure or power violations in the Phase 5 inventory. The observed aggregate violations are therefore enthalpy-dominated. Two S1 rerun files store aggregate rates without the per-constraint breakdown; they are included in aggregate safety statistics and excluded from the per-constraint decomposition statement.

S6 GP Calibration Diagnostics

Table S5 summarizes the held-out GP calibration diagnostics. A scenario-specific GP is trained from $N_{\text{train}} = 500$ transitions and evaluated on $N_{\text{held}} = 200$ held-out transitions. The empirical coverage is the fraction of held-out residual errors satisfying the GP-UCB interval $|\Delta \hat{f}_j| \leq \beta \sigma_{\text{GP},j}$.

Table S5: GP calibration diagnostics on held-out residuals. Coverage is averaged over the three modeled residual dimensions.

Scenario	Mean coverage	Mean σ_{GP}	β
S1: Heat	100.0%	0.104	38.15
S2: Pressure	100.0%	0.142	49.29
S3: Coupled	100.0%	0.101	31.84
S4: Nonlinear	100.0%	0.100	5.73
S5: Valve	100.0%	0.100	5.74
S6: Fuel	100.0%	0.100	5.77

The GP-UCB intervals are conservative in the held-out data, so the empirical limitation observed in the main results is not lack of pointwise GP coverage. The limiting factor is how the chosen margin scale reshapes the feasible QP set under each perturbation structure.

S7 Computation Time Analysis

Figure S1 compares the per-step computation time of the safety-filter pipeline (PPO forward pass, GP inference, and QP solve) against the implemented SLSQP-based NMPC baseline. The comparison is with the implemented research baseline, not with optimized industrial NMPC software.

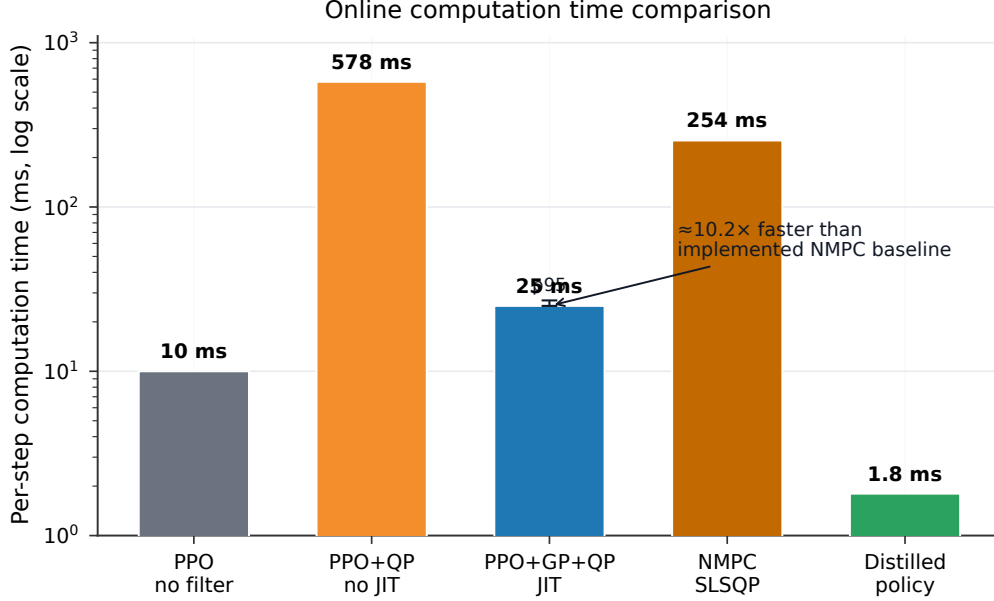


Figure S1: Per-step computation time comparison (log scale). PPO+GP+QP with GPU JIT compilation: ~ 25 ms (p95: 27 ms); NMPC (CPU, SLSQP, $N=5$ horizon): 254 ms average.

S8 GP–CBF Structural Comparison

Table S6 summarizes the structural differences between the proposed robust HOCBF formulation and representative GP-augmented safe-control methods.

Table S6: GP-augmented safety methods compared on structural axes.

Method	Structural axis		
	Uncertainty propagation depth	Per-constraint differentiation	GP-UCB guarantee type
Cheng et al. [1]	Top-level CBF only	No	Per-step CBF ($m=1$)
Hewing et al. [2]	MPC prediction horizon	Partial	Chance-constrained horizon
Ours	Full HOCBF ψ -chain	Yes	Per-step invariance ($m \geq 1$)

S9 Triple-Integrator Validation ($m = 3$)

The compositional robustness margin $\epsilon(x)$ is derived for general relative degree m (Theorem 1 of the main paper), but the CCS constraints have $m \leq 2$. This appendix validates the $m \geq 3$ generalization on a triple integrator system ($n = 3, m = 3$): $\dot{x}_1 = x_2, \dot{x}_2 = x_3, \dot{x}_3 = u$, with a circular keep-out zone $h(x) = (x_1 - c)^2 - r^2$ of relative degree $m = 3$. We evaluate four perturbation types: additive damping ($\Delta f = -\zeta[0, 0, x_2]^\top, \zeta = 0.5$),

periodic ($\Delta f = A \sin(\omega t)[0, 0, 1]^\top$), coupled (damping + periodic), and nonlinear state-dependent ($\Delta f = -\eta[0, 0, x_1 x_2]^\top$). Each method is evaluated over 50 episodes of 500 steps.

Sigma-chain validation A key property of the recursive σ -chain is that uncertainty grows monotonically with the ψ -chain level: $\sigma_1 < \sigma_2 < \sigma_3$. Table S7 validates this across 120 sampled states and all four perturbation scenarios. The ordering $\sigma_1 < \sigma_2 < \sigma_3$ holds for 100% of sampled states, with growth ratios $\sigma_2/\sigma_1 \approx 2.6$ – 2.9 and $\sigma_3/\sigma_2 \approx 2.0$ – 2.2 , consistent with the recursive structure of the σ -chain (Eq. 7 of the main manuscript).

Table S7: Sigma-chain ordering validation ($m = 3$): mean σ_i values and growth ratios across 120 sampled states

Scenario	$\bar{\sigma}_1$	$\bar{\sigma}_2$	$\bar{\sigma}_3$	$\sigma_1 < \sigma_2 < \sigma_3$	σ_3/σ_1
Damping	0.077	0.207	0.437	100%	5.7
Periodic	0.101	0.254	0.514	100%	5.1
Coupled	0.101	0.254	0.514	100%	5.1
Nonlinear	0.077	0.219	0.474	100%	6.1

Safety evaluation Table S8 reports the safety results. The critical result is under the *nonlinear* state-dependent perturbation: HOCBF($m=3$) suffers 13.3% actual constraint violation and 41.5% CBF condition violation, while RHOCBF($m=3$) achieves 0% actual constraint violation with only 3.5% CBF condition violation. The robustness margin $\epsilon(x)$ reduces actual violation from 13.3% to 0%.

Table S8: Triple Integrator Validation ($m = 3$): Constraint Violation Rate (%) under four perturbation types (50 episodes $\times$ 500 steps). “Actual” = entering keep-out zone ($h < 0$); “CBF” = CBF condition violation ($\psi_m < 0$).

Scenario	Method	Actual Viol.	CBF Viol.
Damping	HOCBF($m=3$)	0.00	0.00
	RHOCBF($m=3$)	0.00	0.00
Periodic	HOCBF($m=3$)	0.00	0.00
	RHOCBF($m=3$)	0.00	0.00
Coupled	HOCBF($m=3$)	0.00	0.00
	RHOCBF($m=3$)	0.00	0.00
Nonlinear	HOCBF($m=3$)	13.25	41.45
	RHOCBF($m=3$)	0.00	3.50

The 3.5% CBF condition violation under the nonlinear scenario reflects that the large $\bar{\epsilon} = 20.1$ occasionally renders the QP infeasible, in which case the raw policy action may

violate the CBF condition. Crucially, the system remains a safe distance from the actual constraint boundary, achieving 0% actual violation—the compositional $\epsilon(x)$ provides practical safety even when formal forward invariance cannot be guaranteed due to QP infeasibility.

Compositional vs. constant ϵ The nonlinear scenario provides a setting where $\epsilon(x)$ varies substantially ($\mu_\epsilon = 0.93$, $\max_\epsilon = 3.39$, $\text{std}/\mu = 0.70$). Table S9 reports results across five seeds. All ϵ configurations achieve 0% actual violation, but CBF condition violation rates differ: compositional $\epsilon(x)$ (3.30%) outperforms ϵ_0^{mean} (4.70%) and underperforms ϵ_0^{max} (1.50%)—the compositional $\epsilon(x)$ balances conservatism by being tighter in well-observed states while maintaining protection at the boundary.

Table S9: Triple Integrator ϵ Ablation under Nonlinear Perturbation ($m = 3$, 5 seeds $\times$ 50 episodes $\times$ 500 steps). Mean ϵ is the total per-step robustness margin aggregated over all evaluation steps. The compositional value reflects the accumulated σ -chain contribution entering the $m = 3$ HOCBF inequality, whereas ϵ_0^{mean} and ϵ_0^{max} are constant baseline margins. Results: mean $\pm$ std over 5 seeds.

ϵ Configuration	Actual Viol. (%)	CBF Viol. (%)	Mean ϵ
Compositional $\epsilon(x)$	0.00 ± 0.00	3.30 ± 0.24	21.6
ϵ_0^{mean}	0.00 ± 0.00	4.70 ± 0.24	0.93
ϵ_0^{max}	0.00 ± 0.00	1.50 ± 0.00	3.39
No ϵ ($\epsilon = 0$)	13.43 ± 0.12	41.23 ± 0.07	0

Sparse GP coverage and task performance A position constraint $h(x) = x_{\text{lim}} - x_1$ with constant gradient $|\partial h / \partial x_1| = 1$ combined with sparse GP ($n = 50$ samples confined to $x_1 \in [-0.3, 0.5]$, constraint at $x_{\text{lim}} = 1.2$) creates $17\times$ $\epsilon(x)$ variation. Table S10 shows that compositional $\epsilon(x)$ enables task completion that constant ϵ_0 prevents: $\bar{x}_1 = 0.37$ (31% of distance to constraint) with 0% violation vs. $\bar{x}_1 = -0.38$ and -1.33 for constant margins. Without ϵ , the state reaches $\bar{x}_1 = 0.59$ but with 12.2% violation.

Table S10: Sparse GP Ablation: Position Constraint with Partial GP Coverage ($m = 3$, nonlinear perturbation, 5 seeds $\times$ 10 episodes $\times$ 200 steps). GP trained on $n=50$ samples in $x_1 \in [-0.3, 0.5]$; constraint at $x_{\text{lim}} = 1.2$. Results: mean $\pm$ std over 5 seeds.

ϵ Config.	Viol. (%)	CBF Viol. (%)	QP Infeas. (%)	$\bar{x}_1$	x_1^{max}	Mean ϵ
Compositional $\epsilon(x)$	0.00	0.00	0.0	0.37	0.74	0.15
ϵ_0^{mean}	0.00	0.00	0.0	-0.38	0.00	1.20
ϵ_0^{max}	0.00	0.00	0.2	-1.33	0.00	2.72
No ϵ ($\epsilon = 0$)	12.2 ± 0.0	56.8 ± 0.7	0.0	0.59	1.35	0

S10 Proofs of Theoretical Results

S10.1 Residual Gradient Bound

Lemma 1 (Residual gradient bound). *Let $\delta_i = \psi_i - \hat{\psi}_i^0$ with $\delta_0 \equiv 0$. Under the smoothness and gradient boundedness conditions stated in Theorem 1 of the main manuscript and smoothness of $\hat{f}$ on the compact operating region $\mathcal{X}_{\text{op}}$, $\|\nabla_x \delta_i(x)\| \leq G_\delta^{(i)}$ for all $x \in \mathcal{X}_{\text{op}}$, with*

$$\begin{aligned} G_\delta^{(0)} &= 0, \\ G_\delta^{(i)} &= H_\psi^{(i-1)} \|\Delta \hat{f}\|_\infty + G_\psi^{(i-1)} G_f + (L_{\hat{f},i} + k_i) G_\delta^{(i-1)} + H_\delta^{(i-1)} \|\Delta \hat{f}\|_\infty + G_\delta^{(i-1)} G_f \end{aligned} \quad (\text{S1})$$

where $H_\psi^{(i-1)} = \sup_x \|\nabla_x^2 \hat{\psi}_{i-1}^0\|_{\text{op}}$, $G_\psi^{(i-1)} = \sup_x \|\nabla_x \hat{\psi}_{i-1}^0\|$, and G_f is the gradient bound from the gradient boundedness condition of Theorem 1 of the main manuscript.

Proof. Differentiate δ_i term by term: $\delta_i = L_{\Delta \hat{f}} \hat{\psi}_{i-1}^0 + (L_{\hat{f}} + k_i) \delta_{i-1} + L_{\Delta \hat{f}} \delta_{i-1}$.

(i) $\nabla_x (L_{\Delta \hat{f}} \hat{\psi}_{i-1}^0) = \nabla_x^2 \hat{\psi}_{i-1}^0 \Delta \hat{f} + (\nabla_x \Delta \hat{f})^\top \nabla_x \hat{\psi}_{i-1}^0$. By Cauchy-Schwarz and the gradient boundedness condition of Theorem 1 of the main manuscript: $\|\nabla_x (L_{\Delta \hat{f}} \hat{\psi}_{i-1}^0)\| \leq H_\psi^{(i-1)} \|\Delta \hat{f}\|_\infty + G_\psi^{(i-1)} G_f$.

(ii) $\nabla_x ((L_{\hat{f}} + k_i) \delta_{i-1})$ contributes $(L_{\hat{f},i} + k_i) G_\delta^{(i-1)}$.

(iii) $\nabla_x (L_{\Delta \hat{f}} \delta_{i-1}) = \nabla_x^2 \delta_{i-1} \Delta \hat{f} + (\nabla_x \Delta \hat{f})^\top \nabla_x \delta_{i-1}$, bounded by $H_\delta^{(i-1)} \|\Delta \hat{f}\|_\infty + G_\delta^{(i-1)} G_f$.

Summing yields Eq. (S1). Under the gradient boundedness condition of Theorem 1 of the main manuscript, $G_f = L_{\hat{f}} \|\Delta \hat{f}\|_\infty = \mathcal{O}(\|\Delta \hat{f}\|)$. By induction, $G_\delta^{(i)} = \mathcal{O}(\|\Delta \hat{f}\|)$ at every level, so $|L_{\Delta \hat{f}} \delta_{i-1}| \leq G_\delta^{(i-1)} \|\Delta \hat{f}\| = \mathcal{O}(\|\Delta \hat{f}\|^2) = \mathcal{O}(\rho^2)$. $\square$

S10.2 Full Proof of Theorem 1

Proof of Theorem 1. Step 1 (GP calibration). By the RKHS boundedness and GP calibration conditions stated in Theorem 1 of the main manuscript and the GP-UCB concentration inequality of Srinivas et al. [4], $|\Delta f_j(x) - \mu_{\text{GP},j}(x)| \leq \beta \cdot \sigma_{\text{GP},j}(x)$ for all $x \in \mathcal{X}$ holds with probability $\geq 1 - \delta/n$ per dimension. By union bound over $j = 1, \dots, n$, this holds simultaneously with probability $\geq 1 - \delta$. Condition on this event.

Step 2 (Compositional bound). We show $|\delta_i(x)| \leq \sigma_i(x)$ by induction on i .

- *Base case ($i = 1$):* $\delta_1 = \sum_j (\partial h / \partial x_j) \Delta \hat{f}_j$. By Step 1 and triangle inequality, $|\delta_1| \leq \beta \sum_j |\partial h / \partial x_j| \sigma_{\text{GP},j} = \sigma_1$.
- *Inductive step ($i \geq 2$):* Assume $|\delta_{i-1}| \leq \sigma_{i-1}$. From Eq. (5) of the main text: $|\delta_i| \leq |L_{\Delta \hat{f}} \hat{\psi}_{i-1}^0| + (|L_{\hat{f}}|_{\text{op}} + k_i) |\delta_{i-1}| + |L_{\Delta \hat{f}} \delta_{i-1}|$. Term 1 bounded identically to base case; Term 2 bounded by induction hypothesis; Term 3 bounded using Lemma S1: $|L_{\Delta \hat{f}} \delta_{i-1}| \leq G_\delta^{(i-1)} \beta \|\sigma_{\text{GP}}\|_2 = \sigma_{\text{cross}}^{(i)}$. Summing: $|\delta_i| \leq \sigma_i$.

The cross term is retained at full weight. Under the gradient boundedness condition of Theorem 1 of the main manuscript, it is formally $\mathcal{O}(\rho^2)$.

Step 3 (Total perturbation). Combining the inductive bounds from Step 2 with the coupling-coefficient bound Eq. (S2) yields $|\Delta(x, u)| \leq \sigma_m(x) + \sum_{j=1}^{m-1} c_j \sigma_j(x) + \sigma_{\text{ctrl}}(x) = \sigma_{\text{total}}(x)$. For $\epsilon_\kappa = 1$, $\epsilon(x) = \sigma_{\text{total}}(x)$, hence $|\Delta(x, u)| \leq \epsilon(x)$.

Step 4 (Forward invariance). For u^* satisfying $A_0 u^* \leq b_0 - \epsilon$: $\psi_m^{\text{true}}(x, u^*) = \hat{\psi}_m^0(x, u^*) + \Delta(x, u^*) \geq \epsilon(x) - |\Delta(x, u^*)| \geq 0$. By Theorem 2 of Xiao et al. [2], the safe set $\mathcal{C}$ is forward invariant. $\square$

S10.3 Coupling Coefficient Gap Bound

The coupling coefficient gap is bounded by:

$$\sigma_{\text{ctrl}}(x) = \beta \sum_j \left| \frac{\partial(L_g L_{\hat{f}}^{m-1} h)}{\partial x_j} \right| \sigma_{\text{GP},j}(x) \cdot u_{\max} \quad (\text{S2})$$

where $u_{\max} = \sup_{u \in \mathcal{U}} \|u\|$. On the CCS, $\sigma_{\text{ctrl}}/\sigma_{\text{total}} < 0.15$ for both constraints.

References

- [1] R Cheng, G Orosz, R M Murray and J W Burdick. End-to-end safe reinforcement learning through barrier functions for safety-critical continuous control. In: *Proceedings of the AAAI Conference on Artificial Intelligence*; 2019. p. 3387–3395. doi: 10.1609/aaai.v33i01.33013387.
- [2] L Hewing, J Kabzan and M N Zeilinger. Cautious model predictive control using Gaussian process regression. *IEEE Trans Control Syst Technol* 2020;28(6):2736–2743. doi: 10.1109/TCST.2019.2949757.